parkflow//

Proposal for Sime Darby Property

PROPOSED PARKING SERVICE

Priority parking for SDP malls

Open to every shopper; fee credited for members. KLGCC Mall is a proposed first pilot.

Team Hexa Universiti Malaya, revision 10 November 2026



Leow Shen En
Data Science
TEAM LEAD



Pohyi Chong
Data Science



Isaac Teo
Data Science



Lee Jia Shern
Actuarial Science



Choong Zhuo Lin
Artificial Intelligence



PROPOSED FIRST PILOT
KLGCC Mall

Sources: S1: KLGCC Mall official parking information. Property image: KLGCC Mall exterior, supplied by Team Hexa; original source to be confirmed. Independent Team Hexa proposal.

Parking uncertainty can cost malls planned visits

25 min

Spent searching for parking daily in Kuala Lumpur

BCG for Uber, 2017, as reported

80%

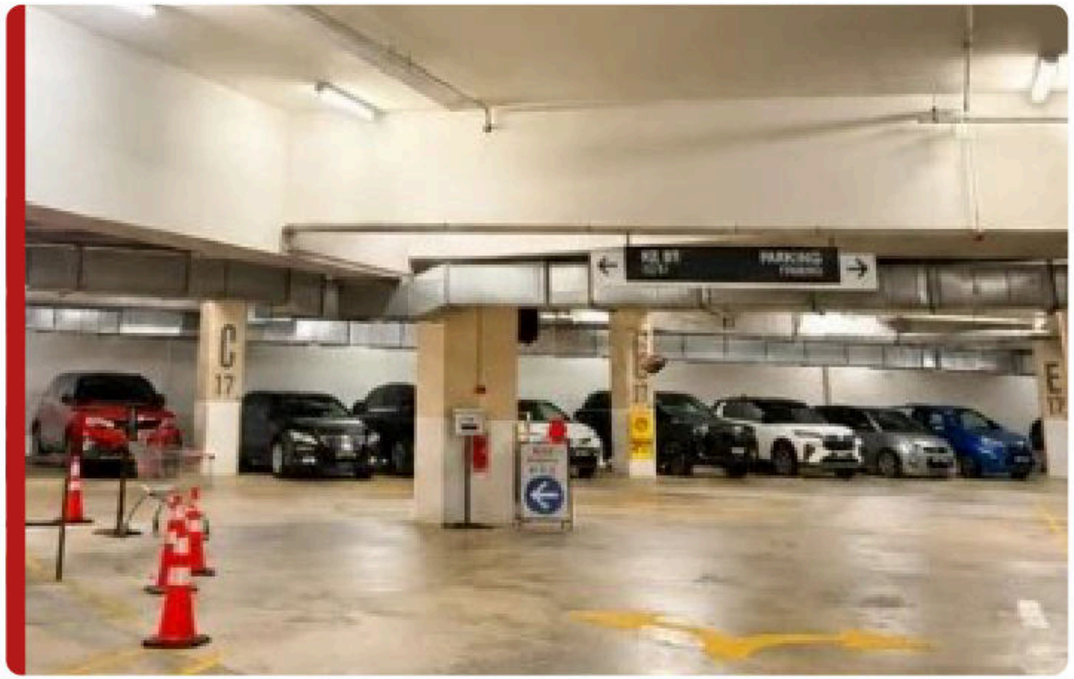
of Malaysian urban trips rely on private vehicles; 87.3% drive alone in Klang Valley

MOT / Malay Mail, 2025; Cent-GPS

2.6 M

users on Smart Selangor Parking, backed by 89%+ Malaysian smartphone adoption

UUM JDSD, 2025; MyGov



KLGCC Mall – site observation

Parking area during our site visit (9 Nov 2026), where most bays were occupied.

Bays are finite, so the question is allocation: parking friction can lead to lost visits, lower spending and weaker repeat visits.

Sources: BCG for Uber (SAYS) · MOT / Malay Mail (2025) · Cent-GPS / NST · UUM Smart Selangor Parking Study (2025). Malaysian market benchmarks, not KLGCC measurements. Site photo: one Team Hexa visit, not a bay-occupancy count.

Existing programmes leave capacity efficiency open

Existing solution	What they offer (publicly available)	Remaining opportunity for ParkFlow
1 Utama ONECARD	Member preferred parking and LPR-linked access/payment	**Reliable advance reservations** Commit and fulfil parking slots for members
Sunway Pyramid PARKEASY	Preferred parking and selected EV bay reservations (via ParkEasy)	**Broader reservation flexibility** Extend beyond EV bays to the whole zone
Central i-City I-CITY APP	LPR ticketless access and cashless payment via the i-City app	**Demand-responsive capacity allocation** Adjust protected capacity based on expected demand

THE OPPORTUNITY

How can malls reliably honour parking reservations without sacrificing walk-in capacity?

Sources: S2: 1 Utama ONECARD official FAQ · S3: Sunway Pyramid official FAQ · S4: TimeTec i-City parking project (vendor page). Capabilities reflect reviewed public pages; unmentioned capabilities are not treated as absent. The opportunity column is proposed ParkFlow scope, not a verified market gap.

A reservation service for one parking zone

Any shopper can reserve a dedicated zone (one level, ~200 bays); members' fee is credited to their parking bill.



Illustrative 3D layout of a whole level. Not a surveyed or approved KLGCC floor plan.

How ParkFlow works

01 **Reserve access**
Book a space in the zone before arrival. Members use their mall membership; the fee is credited to the bill.

02 **Verify at entry**
The existing LPR recognises the plate, confirms the booking and admits the car.

03 **Park in any available bay**
Choose any free bay in the zone. Unclaimed bookings are released after a 15-minute grace period.

Intelligent capacity management — A booking is a claim on the zone, not a fixed bay, so walk-ins keep their space. How the capacity is planned is shown later in this deck.

For shoppers
A more certain, time-saving parking experience.

For SDP
Booking income from non-members, member engagement and tenant sales to test in the pilot.

For the portfolio
Prove the model at one site, then replicate across SDP malls.

Proposed service. A whole level of about 200 bays is the planning size (Team Hexa site visit; level not named). Offline simulations were rerun at 200 bays (appendix). LPR and CCTV are site observations, to be confirmed with the mall. Benefits are pilot hypotheses, not measured results.

RAMP up the car park you already own

From **what the mall has today** to **profit**, in four steps.

Reuse

what the mall already has

Car park, plate cameras, barrier, membership and payment all stay. We add one booking layer on top.

EXISTING · LOW COST

Attract

high-value members

A guaranteed bay near the lifts: free for members, extra for others. Frequent shoppers get a reason to join.

PILOT TESTS THE EFFECT

Monetise

the parking fee

Non-members pay a booking fee. Members' fee is credited to their bill. The fee alone can repay the project.

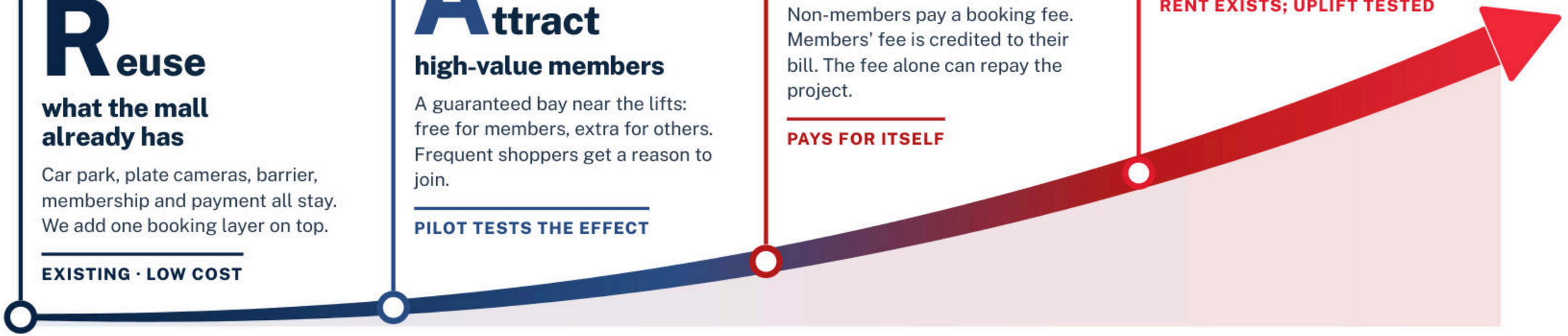
PAYS FOR ITSELF

Profit

as members spend more

Members visit and spend more. Tenant sales rise, and tenants pay SDP a share of their sales as turnover rent.

RENT EXISTS; UPLIFT TESTED

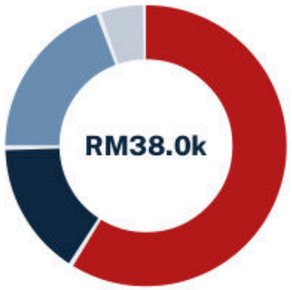


Source: SDP Annual Report 2022 (investment highlights); excerpt on slide 35.

KLGCC zone: RM38.0k upfront (capex), RM6.8k a year (opex)

CAPEX One-time, upfront investment

Display, LPR and hosting prices are listed; the rest are assumptions the site survey and a vendor quote will replace.



Third-party integration and security Vendor interface + gateway and pen test; in-house build not charged	**RM22,500**
Blind-spot CCTV, 4 cameras RM1,500 each; quote after survey	**RM6,000**
Power, network and works Allowance; survey firms it up	**RM7,500**
Guidance display (optional) Viskou list price, tax extra	**RM2,033**
LPR cameras Existing at KLGCC; RM3,226 each if absent	**RM0**

TOTAL UPFRONT

RM38,033

range RM30.0k to RM54.9k (high: 10 cameras, +35% overrun)

Not yet priced gate work and equipment SST.

OPEX Recurring, every year

Hosting is priced; maintenance covers the third-party build only (15% a year). In-house support is not charged.



Maintenance 15% of third-party build a year	**RM3,375**
Hosting 2 servers, AWS Malaysia, 24×7	**RM3,400**

TOTAL RUNNING

RM6,800 a year

maintenance 15% of third-party build, plus hosting

Not yet priced central support.

Third-party items are **59% of upfront cost** and are allowances, not quotes. The in-house build (about **RM55k** at market rate) is not charged: SDP is assumed to have the talent.

Later malls pay site integration only; the platform build is paid once. Sources: Viskou display; Hikvision 4MP; Payscale; AWS Malaysia. Effort days, cable run and labour are assumptions. Who funds the build is not yet agreed: these are zone costs, not a quote. Derivation: appendix.

About 24 bookings a day pay the project back in 3 years

Bookings a day to repay RM38k and run it

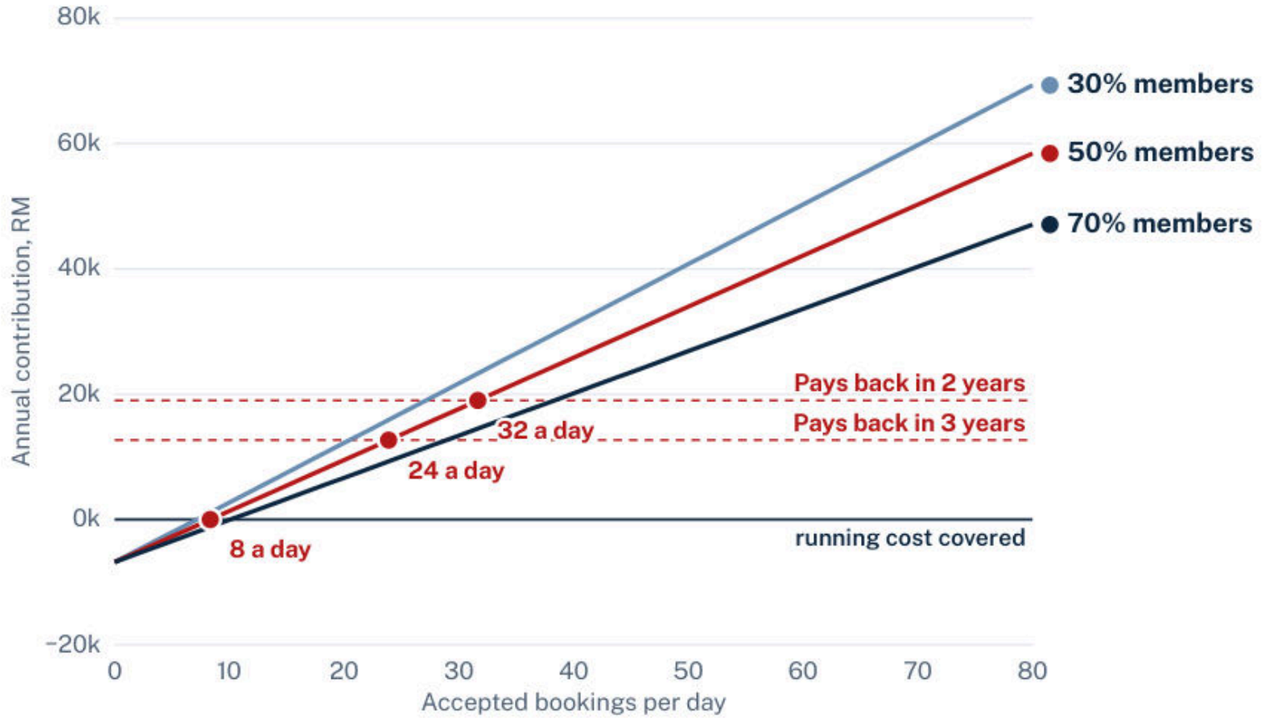
8 **bookings a day**
cover the running cost
about 250 bookings a month

24 **bookings a day**
pay the project back in 3 years
about 717 bookings a month

32 **bookings a day**
pay it back in 2 years
about 950 bookings a month

At 50% members. A non-member pays RM4 (weekday) or RM5 (weekend) a booking; a member's fee is credited to their parking bill.

Annual contribution by volume and member share



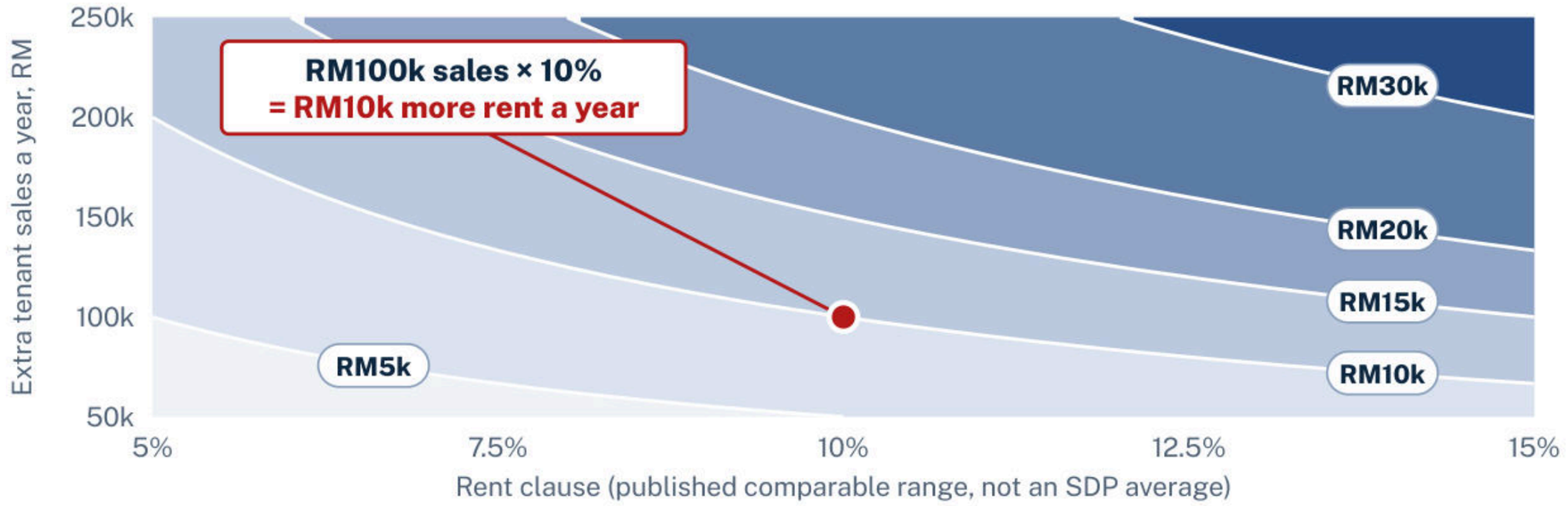
After 6% SST, 20% operator share, card fees and RM6.8k running cost.

Capacity is not the limit: **demand and willingness to pay are**. The pilot measures how many customers book and how many are members.

Contribution = accepted bookings × gross per booking × 365 × net factor − running cost, from the 200-bay simulation. Volume and member share are scenarios; the 20% operator share is unagreed. Fee rules by member type: appendix. Derivation: appendix.

Upside for SDP: each extra member visit adds RM13–39 in rent

Extra rent SDP earns a year = extra sales × rent clause



One extra visit earns SDP **RM13–39** in rent.

SDP already earns turnover rent

Investment & Asset Management

FINANCIAL PERFORMANCE

IAM recorded an 18% increase in revenue to RM107.9 million as compared to FY2021, with the retail business and commercial assets as main contributors. The increase in revenue from retail was mainly due to new tenants opening and a higher turnover rent post COVID-19 lockdowns. Revenue from commercial assets was mainly from higher rental income arising from better occupancy and rental rates, further enhanced by increased income from the car parks.

SDP Annual Report 2022, p.126: higher turnover rent and car park income (highlights added).

The fee pays for the project; this is on top.

Rent range: Tropicana lease terms (SDP's own rate to replace). Trip spend: Vase.ai. Rent income: SDP Annual Report 2022. Extra sales below a tenant's rent floor add nothing.

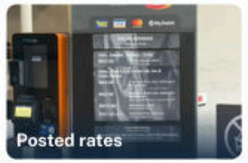
The ask: an eight-week pilot at KLGCC

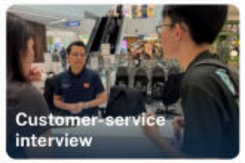
WORKSTREAM AND OWNER		W1–2 Prepare		W3–4 Reserve		W5–6 Shadow AI		W7–8 Decide	
		1	2	3	4	5	6	7	8
Member baseline, price and voucher arms	Loyalty	■	■						
LPR counts, integration, guard readiness	Operator, IT	■	■						
Installed scope and net capture	Finance	■	■	■	■				
Controlled reservations, fixed buffer	Operator, Loyalty			■	■	■	■		
Assisted service and incident log	Operator			■	■	■	■	■	■
Optimizer shadow replay on real events	Team, Operator					■	■		
Cost and investment review	Finance					■	■	■	■
Visits vs control, price and voucher arms	Loyalty							■	■
DECISION MILESTONES			◆ Reservation readiness				◆ Live-AI decision		◆ Scale / extend / stop

SITE VISIT, KLGCC MALL, 9 NOV 2026


Entry barrier and LPR


Aisle camera


Posted rates


Customer-service interview

DECISION GATES

G0 — **Approve scope discovery**
No CAPEX yet

G1 — **Installed quote → investment approval**
CAPEX RM38.0k

G2 — **Readiness check → go live**
Running RM6.8k a year

G3 — **Scale, extend or stop**
Month-6 review, then other SDP malls

THE ASK

Nominate a KLGCC sponsor and operator counterpart, and enable the site survey, integration talks and baseline counts. **Reference upfront RM38.0k** (range RM30.0–54.9k) plus RM6.8k a year. In-house build is not charged (about **RM55k** at market rate).

Proposed pilot sequence and owners. Bars show planned work, not completed delivery or agreed SLAs. Reservation readiness includes reconciled counts, entitlement checks and a signed recovery SOP. Site checks and pass tests: appendix slide 16. Photos: Team Hexa site visit. Site access and baseline instrumentation precede Week 1; live AI activation follows shadow-mode acceptance.

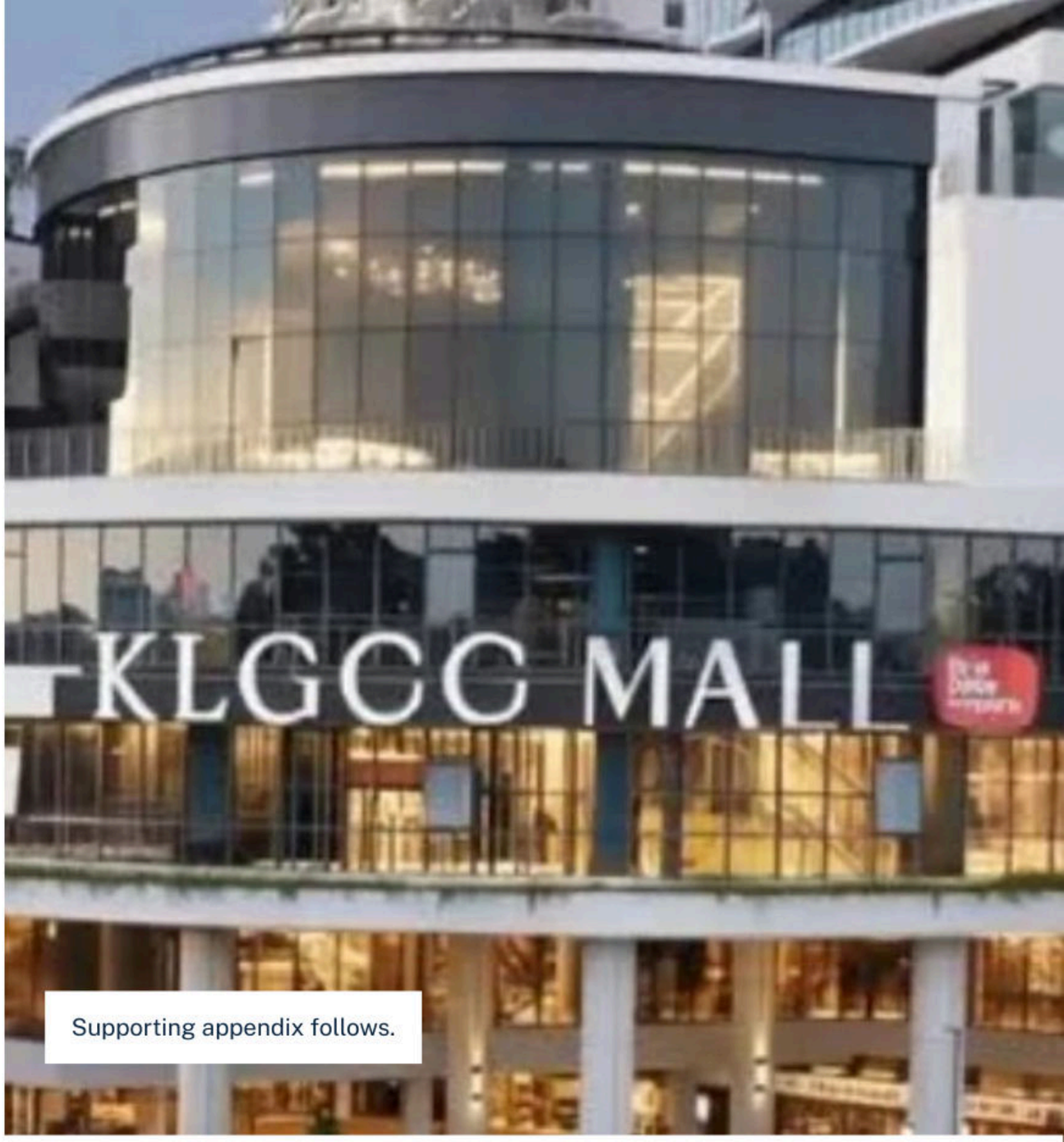
parkflow//

Thank you.

Your parking data may tell you more than you think.

Team Hexa, Universiti Malaya

Questions & discussion



Supporting appendix follows.

Appendix index: click an entry to jump to it

PRODUCT AND PROTOTYPE

12 **Prototype demo**
What is built and what is simulated

13 **Parts and price**
How the booking connects to the mall systems

14 **How capacity is planned**
Plan, hold and release bays over time

15 **Operator view**
Capacity, plans and exceptions

OPERATIONS AND RISK

21 **Exceptions and fault handling**
Uncertain counts, refunds, data incidents

PLAN

16 **Site readiness**
Three site checks before any capex; photos

31 **Implementation roadmap**
Phases and spend

32 **Three-horizon roadmap**
One site to external operators

33 **Pilot measures**
Metrics and owners

TECHNICAL DESIGN

17 **Architecture: four layers**
What is tested and what is proposed

18 **How the planner decides**
Accept rule, stay model, safeguards

19 **Baseline buffer rule**
The simple rule the planner is compared with

20 **Edge cases and unseen events**
Detect, contain, fall back, learn

22 **Occupancy forecasting**
28 UK car parks, six weekly holdouts

COMMERCIAL AND COST

28 **Commercial and value derivations**
Net factor, break-even, value to SDP

29 **Software and running cost**
Third-party items, in-house value, running

30 **Deployment options**
Upfront, running, payback by option

EVIDENCE

23 **Parking policy comparison**
Planner against simple rules

24 **Booking failures vs walk-ins**
What the booking guarantee costs

25 **Robustness to assumptions**
60 random worlds

26 **200-bay zone simulation**
Capacity, grace period, count drift

SOURCES AND DATA

27 **Data register**
Sourced, simulated or assumed

34 **Benchmark methods and scope**
What is real data

35 **SDP Annual Report excerpt**
Turnover rent and car park income, FY2022

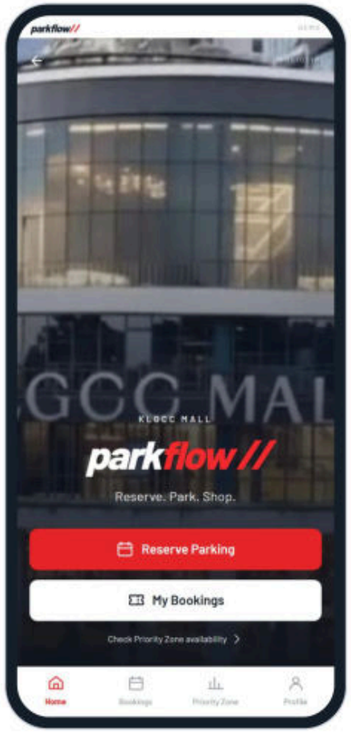
36 **Public sources**
Links and checks

37 **Cost sources**
Price listings and rates

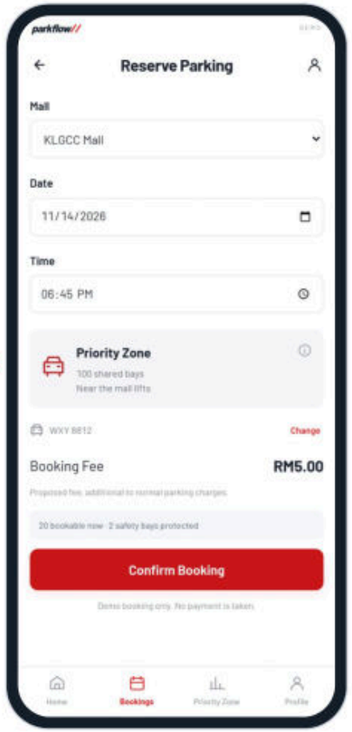
Slide numbers are positions in this deck. Each appendix slide has an index link in its footer.

The prototype: what is built, what is simulated

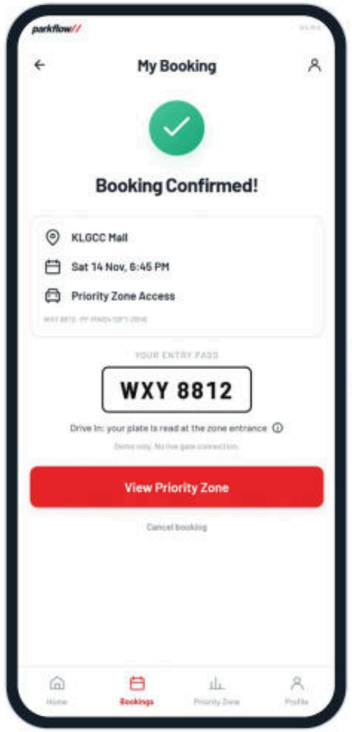
1. HOME



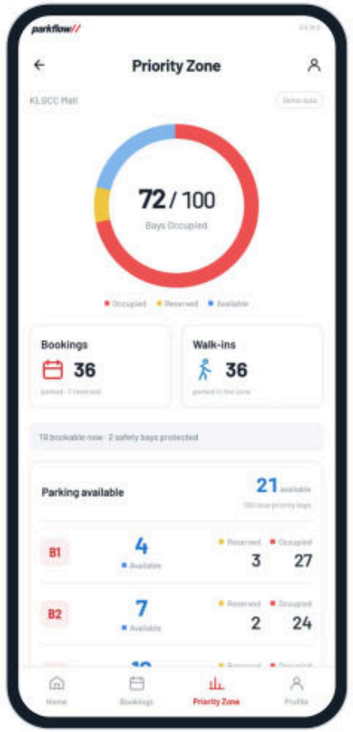
2. RESERVE



3. CONFIRM



4. ZONE



What exists today

Built

A booking flow: mall, date and arrival time, with the fee and no-show rule shown before confirming.

Simulated

The plate-based entry and the LPR check. No live gate, payment or controller connection.

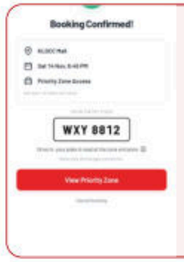
Zone view

Occupied, reserved and available bays by level, with safety bays protected (simulated data).

Screens captured from the ParkFlow driver prototype, simulated data (demo bay counts and a RM5 demo fee, not the proposed zone size). No payment processing or live gate connection. Offline simulation results are separate.

Behind the booking: the parts and the price

CUSTOMER SIDE



New
Booking app
Book; the plate is the pass.

Existing
Mall membership
Identity and member rate.

Existing
Payment
Fee by card or e-wallet.

ParkFlow controller

New

1 **Booking ledger**
claims and no-shows

2 **Stay forecast and planner**
re-plans every 15 minutes

3 **Safety guard**
verified free bays decide

4 **Price and credit rules**
fee credited to a member's bill

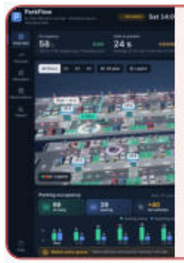
ON SITE



Existing
LPR and barrier
Reads plates, admits, counts.



Existing
CCTV
Audit only; covers blind spots.



New
Operator dashboard
Plan, verify, audit.

WHAT THE CUSTOMER PAYS PER VISIT

Booking fee RM4 weekday, RM5 weekend; kept on a no-show. Slashes show weekday / weekend. Weekday parking is RM3 at any length.

Stay	Normal parking rate	Member booking	Non-member booking
1–3 hours	RM3	**RM4 / 5**	RM7 / 8
Weekend 4 h	RM5	**RM5**	RM10
Weekend 6 h	RM9	**RM9**	RM14
Weekend 7 h+	RM10	**RM10**	RM15

ANOTHER WAY IN: EARN A BOOKING

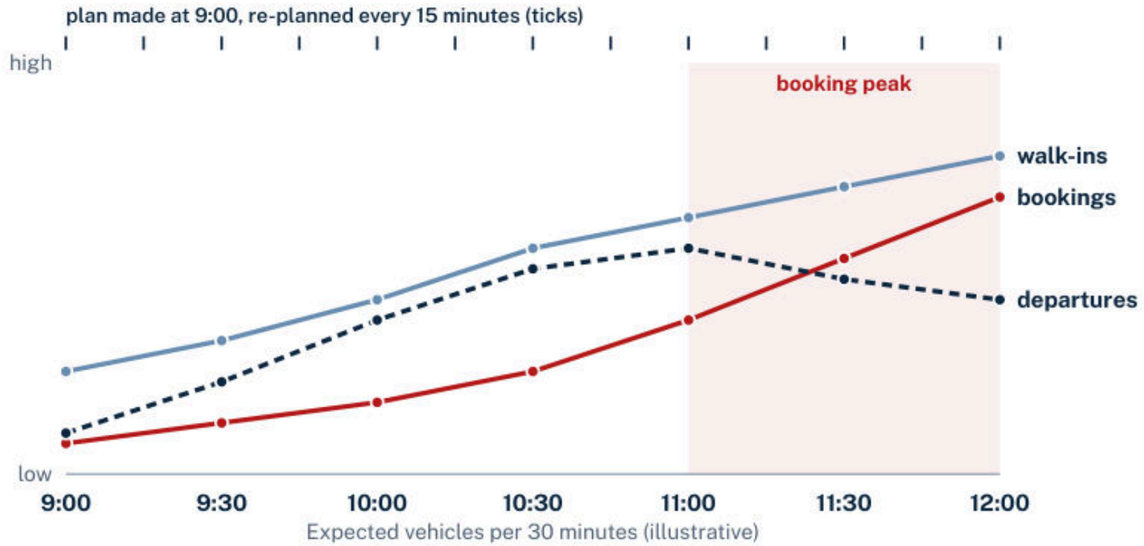
EARN Spend above a set amount in store, get one booking.
COST Fee forgone; pays back if 1 in 3 to 1 in 8 visit again.
PILOT Tested against the paid booking. Store-issued tickets later.

THE RULE A member never pays more than the **larger of the fee and the parking bill**. A non-member pays **both**. That gap, RM3–10 a visit, is the reason to join.

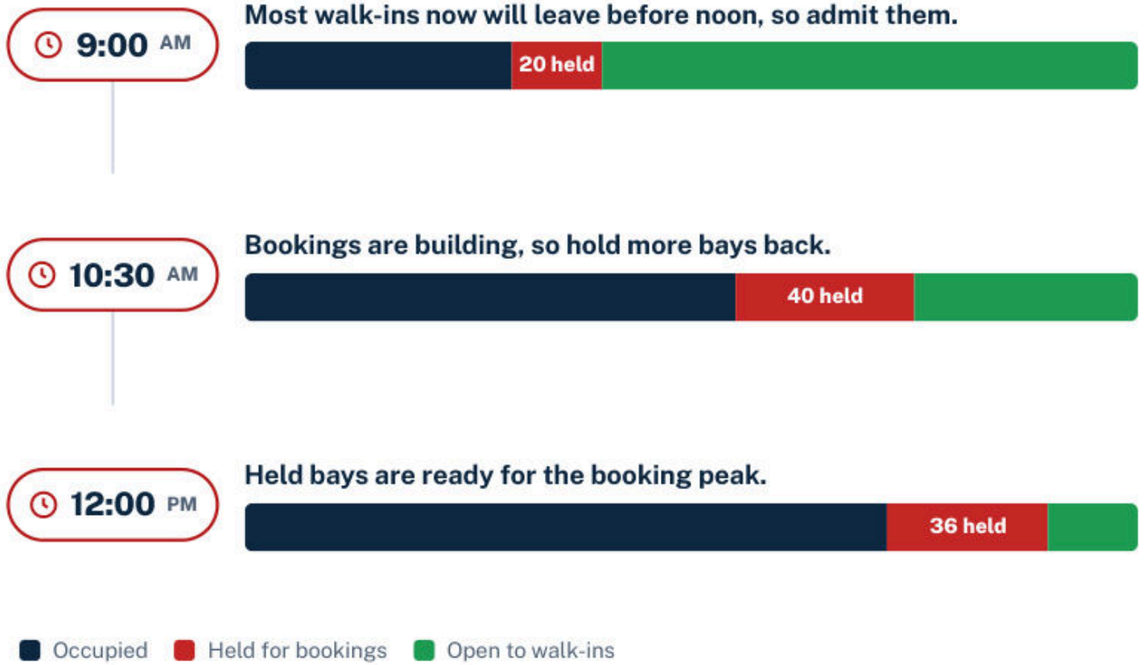
KLGCC tariff: RM3 weekday; weekend RM3 for the first 3 hours, then RM2 an hour, capped at RM10, excluding 6% SST (official page). Weekend hours round up. Booking-service tax is unconfirmed. The member credit is a proposed rule, not an SDP policy. LPR and CCTV are site observations, to be confirmed with the mall.

A booking is a commitment, planned ahead

What the plan looks at



How the 200 bays are split over time illustrative



1 Booking ledger
Confirmed bookings are hard constraints.

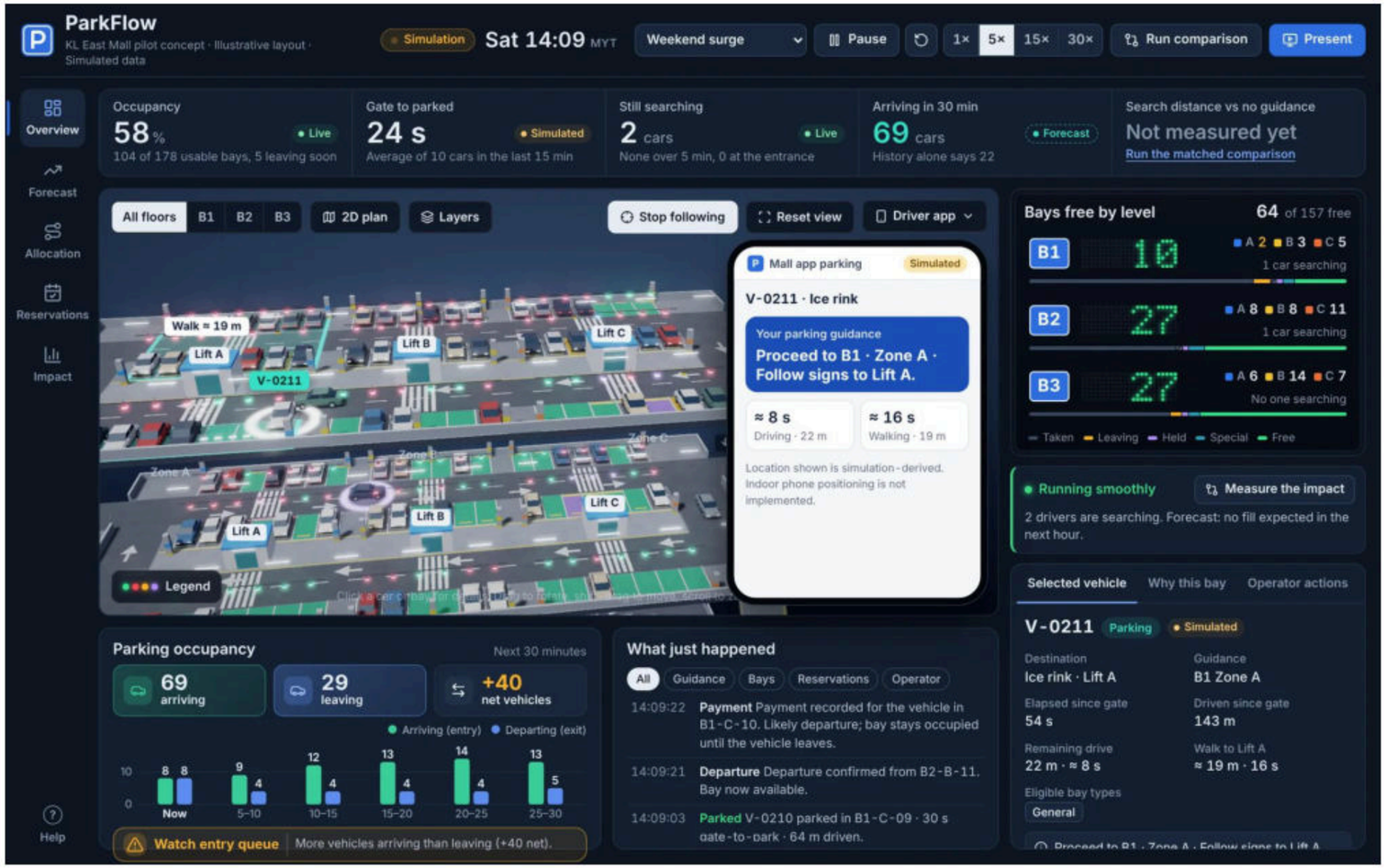
2 Stay forecast
Arrivals, departures and overstay risk.

3 Rolling planner
Re-plans every 15 minutes; three-hour lookahead, widened if stays run long.

4 Safety guard
Verified free bays and plate counts decide every admission.

Illustrative design, not measured. A booking is a claim on capacity, not one named bay; confirmed bookings are planned as hard constraints and verified free bays decide every admission. An overstaying car can still break a booking, so the pilot measures failed bookings per 100, walk-ins turned away and idle bay-hours against a simple buffer rule. Simulation evidence and limits: appendix.

One operator view of capacity and exceptions



Existing prototype view. Rolling-horizon recommendations and policy comparisons are proposed additions.

Capacity, plans and exceptions

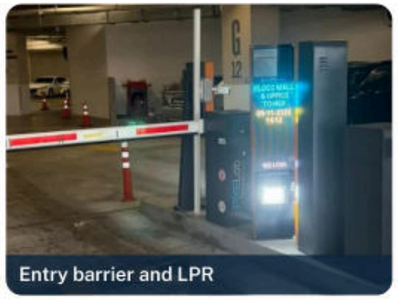
Plan	Review the planner's recommendation and its uncertainty.
Verify	Check current counts, protected promises and the admission guard.
Revenue	Track paid vs credited bookings, no-shows and voucher use (proposed addition).
Audit	Log plan changes, overrides, rejected recommendations and recovery.

Assistance and emergency egress remain available during faults.

Three site checks come before any capex

SITE VISIT AT KLGCC MALL, 9 NOV 2026

Team Hexa on site: photos and a customer-service interview. **No capex before gate G1;** if a check fails, scope shrinks first.


Entry barrier and LPR

01 Zone counts from LPR

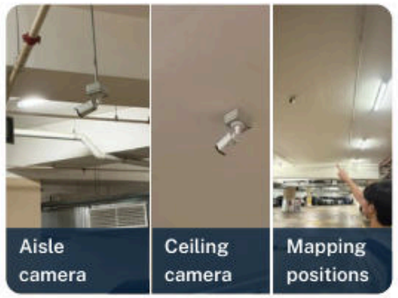
OBSERVED 9 NOV

LPR at the entry barrier

CONFIRM IN WEEKS 1–2

- **Data and API access** — Vendor, API or export
- **Read accuracy** — Share of plates read correctly
- **Zone count** — Entry and exit events, plate ledger

PROPOSED PASS TEST Counts match a manual count over one full day.


Aisle camera | Ceiling camera | Mapping positions

02 Camera coverage

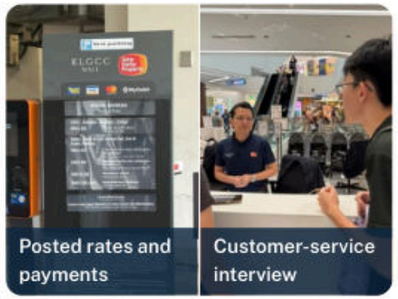
OBSERVED 9 NOV

Aisle and ceiling cameras

CONFIRM IN WEEKS 1–4

- **Blind spots** — Coverage gaps
- **Footage access** — Access and retention period
- **Glare and pillars** — Light and obstructions

PROPOSED PASS TEST Every aisle covered, or gaps mapped and priced.


Posted rates and payments | Customer-service interview

03 Member and payment systems

OBSERVED 9 NOV

Card and e-wallet payment, posted rates

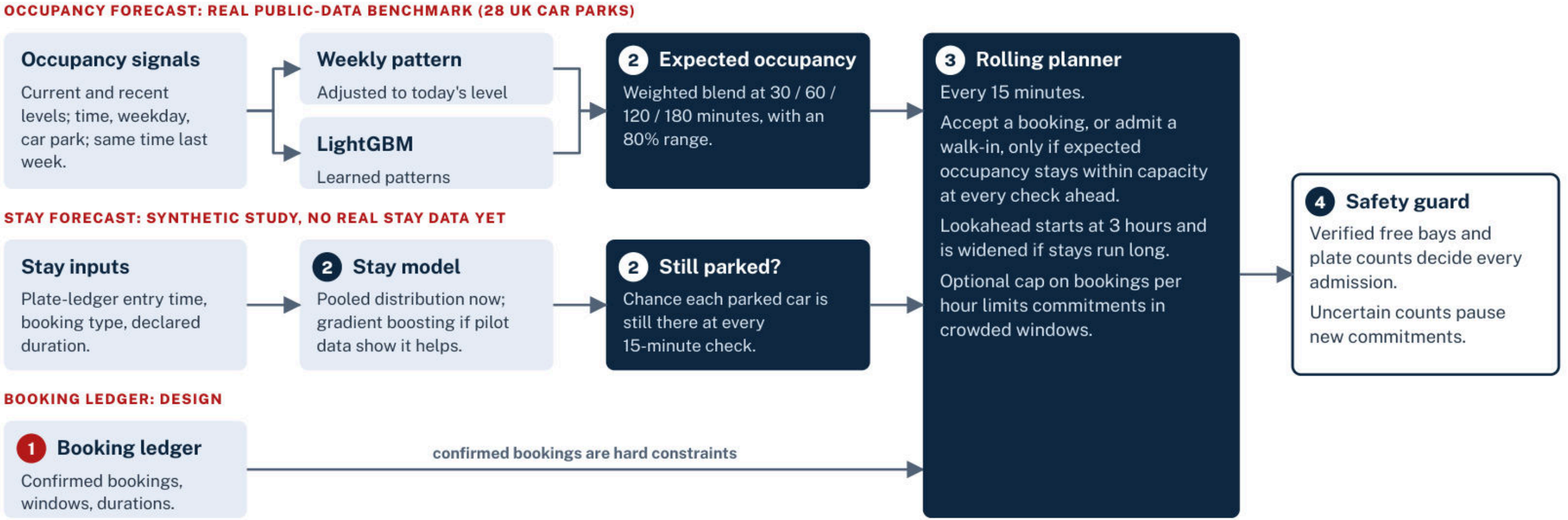
CONFIRM IN WEEKS 1–2

- **Member API** — Membership identity
- **Settlement** — Refunds and no-shows
- **Booking expiry** — Link to the controller

PROPOSED PASS TEST Member lookup, payment and refund work in a test account.

Sources: KLGCC Mall: official parking information. Observed items are Team Hexa site-visit observations (9 Nov 2026), not mall-confirmed. Confirm controller, power/network, gate safety, local coverage and incumbent interfaces.

Architecture: four layers and what is tested



Numbers 1 to 4 mark the four layers; the proposed site loop is log actuals, check errors, retrain, validate before deployment. Weather and event inputs did not improve the forecast on 28 UK car parks, so they are not used. Site connections remain proposed.

Sources: occupancy forecast code; stay model and planner simulation; weather and event test. Diagram redrawn for the commitment architecture.

How the planner decides

Booking request

Accept if, at every 15-minute check during the booking's stay window, expected occupancy plus this booking stays within capacity.

- **Expected occupancy** = parked cars still expected to be there + confirmed bookings, each counted as certain to arrive.
- **Safety margin** = z × the spread of that estimate. z is set separately for bookings and walk-ins.
- An optional cap on bookings per hour limits commitments in crowded windows.

Walk-in arrives

Admit if a bay is physically free and the same check holds at every 15-minute point inside the lookahead, using that car's expected stay.

- Later walk-ins face the same check, so protecting bookings needs no separate demand forecast.
- The lookahead starts at 3 hours; the pilot tests longer ones (5 to 8 hours) against observed stay lengths.

Example: a 7 PM booking made at 9 AM

9:00 accept only if capacity for 7 PM can be committed. 12:00 admit walk-ins unless a long stay could still be parked at 7 PM. 16:00 tighten as long stays begin to overlap. 18:00 confirm free bays. 19:00 honour the booking.

Stay model (layer 2)

- **Inputs:** time of entry, how long the car has been parked, booking type and declared duration; plate history where available.
- **Output:** the chance each parked car is still there at each check. It starts as one pooled stay distribution; gradient boosting is added only if pilot data show it helps.
- **Refresh:** every 5 to 15 minutes from the plate ledger.

Safeguards

- Verified counts decide admission; the plan only adds restrictions.
- Uncertain counts pause new commitments (see fault handling).
- The simple buffer rule on the next slide runs alongside in shadow mode as the comparison.

Settings the pilot calibrates

Lookahead, the two margins, the booking cap, and the planned show-up rate (1.0 means no overbooking).

Design specification. The rule and settings come from the simulation study in this appendix; none is calibrated to KLGCC data. Stay-model inputs other than entry time and age are proposed.

Baseline rule: a buffer that follows demand

The rule, refreshed every 15 minutes

B = clamp(requests per hour × hold time , floor , cap)

- **Requests per hour:** booking requests in the last 30 minutes, doubled and smoothed.
- **Hold time:** about 45 minutes, how long a held bay waits to be claimed.
- **Floor and cap:** 2 bays, and 10% of the zone (20 bays for a 200-bay zone).
- **Free bays** = total − occupied − accepted claims − safety bays.

Worked example: 200-bay zone, B = 6

Free bays	Walk-in arrives	Booking request arrives
16	Admitted (16 > 6)	Accepted
7	Admitted (7 > 6)	Accepted
6	Turned away	Accepted
0	Turned away	Rejected

B at three demand levels (illustrative)

Demand	Requests, last 30 min	Calculated	Buffer
Quiet	2	3	3
Normal	4	6	6
Surge	14 or more	21	20 (cap)

What a forecast changes

Only the input. It replaces last-30-minute requests with expected requests for the next hours, so the buffer can rise before a surge. In the offline study this did not beat reacting to recent traffic, so the pilot tests it in shadow mode against this rule.

Measured in the pilot

Booking rejection rate, idle bay-hours (free bays while a walk-in is turned away), and walk-ins admitted. The floor, cap and hold time are calibrated on the baseline counts.

Rule and numbers are a design proposal; the demand levels are illustrative. Evidence: event-driven report and 200-bay simulation (synthetic demand).

Unfamiliar events: live counts overrule the model

1 DETECT
Notice the model is out of its depth

- **Arrival-rate shift:** a 30-minute sliding window of booking requests, smoothed, re-sets the buffer every 15-minute control cycle.
- **Residual drift (out-of-distribution flag):** actuals outside the 80% prediction interval for k consecutive checks; a CUSUM test on forecast residuals ASSUMED. Nominal coverage is 79.8% on holdout.
- **Sensor drift:** LPR miss rate and plate-ledger reconciliation errors (ghost and unseen cars, p99) raise the uncertain-count flag.

2 CONTAIN
Commit less, automatically

- **Admission control:** an hourly booking cap limits commitments in a crowded window. Removing it cost up to 58 score a day in the stress test SIMULATED
- **Chance-constrained margin, z × σ:** expected occupancy plus z standard deviations must fit capacity at every 15-minute check; a wider σ accepts fewer bookings.
- **Clamped reserve:** the walk-in buffer B = clamp(rate × hold time, floor, cap) rises to its 10% cap.

3 FALL BACK
Never trust a forecast over a count

- **Count-gated admission:** verified counts decide entry; the plan is a monotone restriction and can never relax the gate.
- **Fail-safe mode:** uncertain counts pause new commitments; assisted verification, egress preserved, incident logged.
- **Shadow-mode baseline:** the simple buffer rule runs beside the planner as the champion-challenger comparison and the fallback ASSUMED

4 LEARN
Fix it before the next one

- **Closed loop:** log actuals, track error, retrain on a rolling window and validate on a holdout before deployment.
- **Failure remediation:** fee refunded, nearest-level bay at no extra charge, failure logged.
- **Recalibration:** failures per 100 bookings are reported, and z, the cap and the stay model refit on observed stays.

STRESS TEST SIMULATED

60 random worlds redrawing pulse size, stay shock, volume and overstay noise; settings not retuned. Hybrid planner scored higher in 93 to 100% of worlds.

WHY EVENTS ARE NOT A MODEL INPUT SOURCED

Weather and event covariates added no accuracy across 28 UK car parks (rolling-origin holdouts), so the design reacts to live state instead of forecasting the event.

NOT YET SHOWN

A real unseen event at KLGCC. If stays are worse than assumed, failures can exceed the ceiling; the pilot measures this and recalibrates.

A surprised model can only make the zone more cautious, never less: verified counts decide every admission.

Sources: simulate_mc.py (60-world stress test, ranges fixed before the run); weather and event test on 28 UK car parks; planner and buffer slides in this appendix. Triggers marked Assumed are proposals whose thresholds the pilot sets; the 200-bay simulation uses synthetic demand.

Exceptions and fault handling

Can the operator trust the current state?

Verified plate, current counts, live controller, valid entitlement

Yes

Apply the capacity guard

Admit eligible vehicles only when verified free bays and the planner's booking checks pass.

Owner: parking operator

No / uncertain

Pause new commitments

Use assisted verification, preserve egress, log the incident and reconcile before restarting.

Owner: operator + IT / integrator

Member exceptions

Published grace, no-show, overstay and refund rules.

Data incident

Restricted access, evidence preservation and SDP privacy escalation.

Booking cannot be honoured

Proposed rule: refund the fee, offer a bay on the nearest level at no extra charge and log a failure. Failures per 100 bookings is a pilot metric.

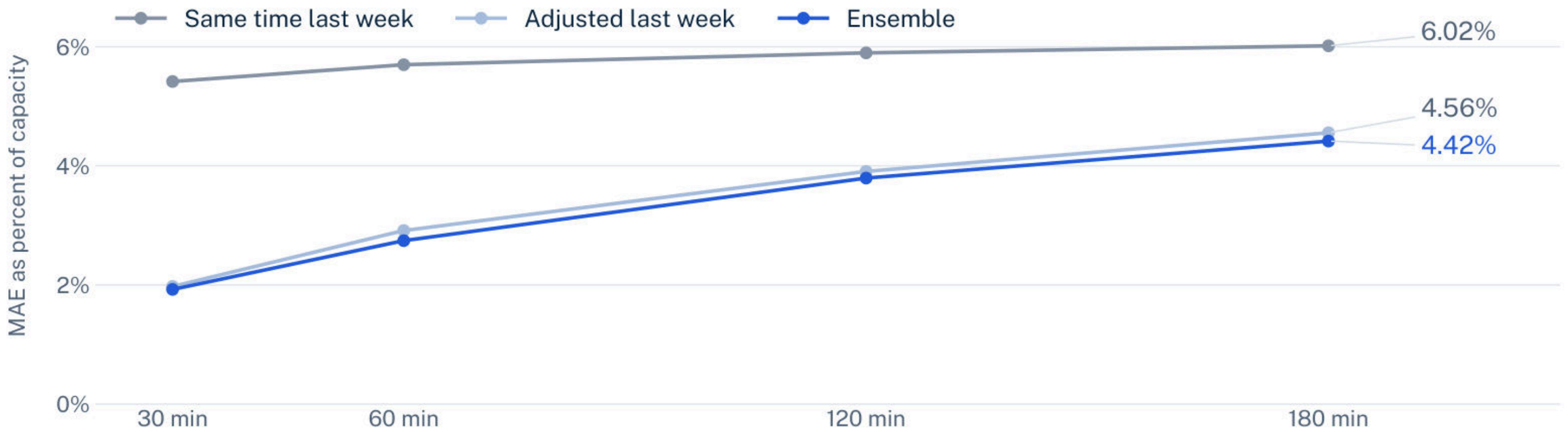
Low paid demand

Review scope and price with finance before expansion.

A booking we cannot honour is refunded and logged as a failure; the pilot reports failed bookings per 100.

Sources: S8: Malaysia PDP official Act/guidelines index. Deployment obligations depend on the actual processing and operator arrangements; this is an implementation workstream, not a legal certification.

Occupancy forecasting



Public occupancy: 18,074 forecasts, 28 UK car parks, six weekly holdouts. Weather and event flags added no accuracy.

Local scope: request and turnover tests use synthetic data. Birmingham occupancy does not validate KLGCC booking demand.

Source: Causal occupancy benchmark; weather and event test. MAE is percent of capacity. Strong adjusted baseline retained; 180-minute nominal 80% interval coverage is 79.8%.

Parking policy comparison

Interleaved synthetic requests, 100 bays, 60 held-out days, selected comparators



Member access and walk-ins trade off. The calibrated fixed buffer serves more member arrivals; the learned-forecast controller admits more walk-ins.

ML versus adaptive forecasts: +0.12 preference-score units/day; 95% interval -2.17 to +2.30. No clear advantage in this comparison.

Source: Event-driven benchmark, 8 Nov 2026. Score = 5 × members served + walk-ins; this is an operating preference, not money. Demand/cue relationships are synthetic assumptions. Full ten-policy results and earlier batch-study null are in the appendix.

Simulation: failed bookings against walk-ins

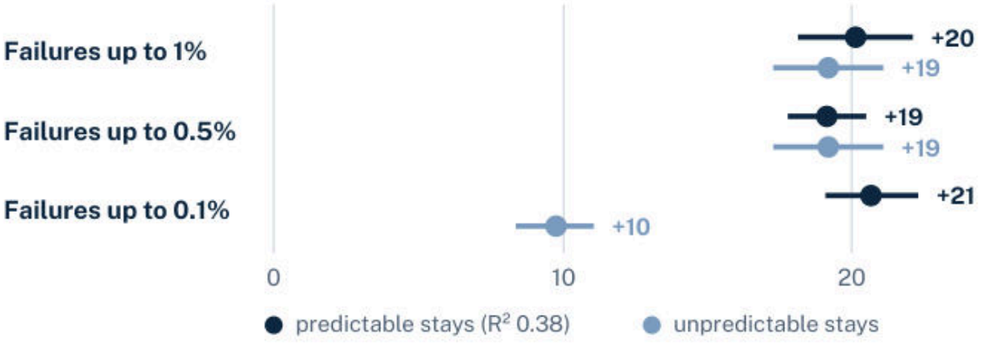
Policy	Bookings honoured / day	Walk-ins / day	Failed bookings	Score	Score vs simple rule
Failure ceiling: Up to 1%					
Tuned simple rule	110	437	0.07%	988	–
Hybrid planner	108	467	0.39%	1008	+20.1 [+18.1, +22.1]
Failure ceiling: Up to 0.5%					
Tuned simple rule	110	437	0.07%	988	–
Hybrid planner	110	458	0.12%	1007	+19.1 [+17.8, +20.5]
Failure ceiling: Up to 0.1%					
Tuned simple rule	110	424	0.04%	976	–
Hybrid planner	110	448	0.00%	997	+20.6 [+19.1, +22.3]

100 bays, 200 fresh test days, settings chosen on 150 other days. Score = 5 × bookings honoured + walk-ins admitted. Brackets: 95% interval. **Limits:** synthetic demand and stays with assumed stay effects; hybrid tested with an 8-hour lookahead only; overstaying cars cause the remaining failures; fallback level and refund not modelled; 173 settings searched on calibration days, test days never used to choose.

What it shows

- The **hybrid** keeps the simple rule's cap on bookings and adds the stay-risk check on walk-ins. It beat the tuned simple rule at every failure ceiling, by about 19 to 21 score a day (1 to 2%).
- It honours about the same bookings and admits **more walk-ins**, at 0.00 to 0.42 failed bookings a day.
- With unpredictable stays the gain shrinks (+10 a day at the strictest ceiling).
- Planning without the booking cap lost by up to 58 score a day at tight ceilings. A last-minute hard hold added nothing in an earlier run.

Hybrid minus tuned simple rule: score per day, 95% interval



Keeping the simple rule's booking cap and adding the stay-risk check on walk-ins gave about 1 to 2% more score at near-zero failures.

Sources: report_v5.md, protocol, simulate_v5.py. Earlier runs, including the first full run (not breach-matched), are kept in the same folder.

Robustness: 60 random sets of assumptions

Policy	Beats simple rule (worlds)	Median score gain [10th, 90th pct]	Failures under ceiling (worlds)	Failure rate: median / 90th pct	Both: higher score and under ceiling
Failure ceiling 1%: the tuned simple rule stays under it in 95% of worlds					
Hybrid planner	100%	+19.5 [+11.2, +29.1]	82%	0.50% / 1.50%	82%
Planner, separate margins	62%	+10.0 [-37.9, +43.7]	72%	0.37% / 1.77%	40%
Failure ceiling 0.5%: the tuned simple rule stays under it in 78% of worlds					
Hybrid planner	97%	+17.5 [+8.2, +25.2]	77%	0.17% / 0.84%	73%
Planner, separate margins	53%	+2.0 [-44.6, +30.9]	75%	0.18% / 0.91%	37%
Failure ceiling 0.1%: the tuned simple rule stays under it in 55% of worlds					
Hybrid planner	93%	+17.5 [+2.8, +30.1]	77%	0.00% / 0.34%	70%
Planner, separate margins	30%	-66.5 [-158.0, +35.5]	82%	0.00% / 0.24%	18%

What it shows

- The hybrid scored higher than the tuned simple rule in **93 to 100%** of worlds, with a median gain of 18 to 20 score a day.
- Failures stayed under the ceiling in 77 to 82% of worlds; the simple rule managed 95% at the 1% ceiling and 55% at 0.1%.
- If real stays are worse than assumed, failures can exceed the ceiling; recalibrate the margins on pilot data.

How it was run

- 60 random worlds, 30 days each, redrawing show-up, stay shock and spread, booking volume, pulse size, overstay noise, stay predictability and advance share. Settings tuned on the base case are **not retuned**; the planner refits its stay model in each world.

Share of 60 random worlds: hybrid planner



grey tick: share of worlds where the tuned simple rule stays under the same ceiling

It scores higher in almost every world, but the failure ceiling must be recalibrated on pilot stay data.

Sources: report_v6.md, simulate_mc.py; ranges fixed in the protocol before the run. The ranges are my assumptions: they test sensitivity and do not make the numbers true for KLGCC.

200-bay zone: capacity, grace period and LPR count drift

Volume sweep: guard + fixed buffer 10, grace 15 min

Requests/day	Accepted	Arrived	Unclaimed	Walk-ins	Rejected
25	22	19	4	938	0.0%
50	44	36	7	912	0.0%
90	79	65	14	867	0.0%
140	122	100	21	820	0.0%
200	176	145	31	743	0.2%

Mean of 40 synthetic days per row. Walk-in demand scales with zone size. Demand shape is assumed.

Grace period at 90 requests/day

Grace	Idle bay-h/day	Late, admitted	Late, denied
10 min	52.3	8.2	0.0
15 min	52.3	5.3	0.0
20 min	52.5	4.2	0.0
30 min	53.3	1.5	0.0

LPR count drift: plate ledger, 5h ghost timeout

Missed reads	Unseen cars (p99)	Ghost cars (p99)
0.5%	5	7
1.0%	6	10
2.0%	10	15
3.0%	11	20

Capacity is not the limit at this scale: demand and willingness to pay are.

Synthetic demand; late-arrival mix 80/12/6/2% is an assumption. Soft release lets a late customer in when capacity exists. Unseen cars (missed entry) undercount; ghost cars (missed exit) overcount, the safe direction. Code: analysis/v3_zone200.py; results: results.json.

Data register: what is sourced, simulated or assumed

Input	Tier	Basis
Reservation fee RM4 weekday / RM5 peak	ASSUMED	Price test. Local anchor: Sunway ParkEasy reservations reported at 6–8 credits (RM1 each) per booking; verify current price.
Standard parking bill	SOURCED	KLGCC official tariff: RM3 weekday; weekend RM3 to 3h, then RM2/h, cap RM10.
Member offset rule	ASSUMED	Fee credited to the bill; excess kept as revenue; deposit kept on no-show. Our design choice.
No-show rate 5–20%	ASSUMED	Other settings report 10–31% (campground model, car rental, one city system). Not parking at malls.
Grace period 15 min, soft release	SIMULATED	Tested at 10–30 min; with soft release the choice barely moves idle bay-hours.
Bookings per day, member share	ASSUMED	No local data. Scenario ranges; the pilot measures both.
LPR and CCTV at KLGCC	SOURCED	Team Hexa site visit, 9 Nov 2026. Vendor, data access and blind spots unconfirmed.
CCTV and site works	ASSUMED	RM1,500 a camera = RM750 camera (sourced) + RM400 cable (RM100 per 10 m, 40 m assumed, unverified) + RM350 labour (half-day, assumed). Power, network and works RM7,500 is a flat allowance for switch, UPS, trunking and signage; no survey yet.
LPR miss rate 0.5–3%	SIMULATED	Count-drift simulation; real read accuracy is a pilot measurement.
Third-party software items RM22.5k, running RM6.8k/yr	ASSUMED	Vendor interface and gateway/pen-test allowances; no quotes yet. In-house build (89 person-days × RM623 = RM55.1k at market rate) is not charged. Running = 15% of third-party build + hosting. Appendix.
Service tax on the booking fee	SOURCED	Parking is taxed at 6% (Customs guide). A reservation fee is not listed as exempt; treated as tax-inclusive: conservative.
Willingness to pay	ASSUMED	Directional only: ParkMobile 67% (US). No local test; first pilot measure.

Order of preference: web sources, then simulation, then assumptions as a last resort. Web-derived items come from search results and must be opened and checked before submission. Nothing here is a measured KLGCC demand or price response.

How the commercial and value numbers are derived

Commercial model (slide 7)

- **Fee per booking:** non-member or no-show RM4 weekday / RM5 weekend; member up to 3 hours RM1 / RM2; member over 3 hours RM1 / RM0. A member's fee is credited to the parking bill, so only any excess counts.
- **Net factor 0.735 per RM of fee:** 0.8 ÷ 1.06 (the operator takes 20% of the ex-tax fee) − 1.944% (card fee on the amount collected).
- **Service tax 6%:** parking is taxed at 6% (Customs guide). A reservation fee is not listed as exempt, so the fee is treated as tax-inclusive. If untaxed, contribution is about 6% higher.
- **Gross per booking:** RM3.04 at 50% members, from the 200-bay simulation (40 synthetic days per volume; deposits credited to members' bills).
- **Displaced parking revenue:** cars sent to other levels pay the same KLGCC tariff, so only walk-ins who leave the mall are a loss. The pilot measures it.
- **Break-even at 50% members:** running cost at 8 bookings a day, 3-year recovery at 24, 2-year at 32; bookings a month = daily bookings × 30.

Upside for SDP (slide 8)

- **Rent per extra visit:** RM260 mean recalled trip spend × 5–15% gross turnover-rent clause (Tropicana valuation) = RM13–39. Turnover rent has a floor: extra sales below it add zero.
- **Voucher:** RM4.6 fee forgone per booking (RM23,430 gross ÷ 5,077.5 paid bookings in the reference mix; an upper bound, as members' fee is credited) ÷ (RM260 mean recalled trip spend × rent clause).
- **Spend gap:** RM195 against RM327 recalled trip spend (Vase.ai, N=500), +68%. Association only.
- **Pilot measures:** member sales versus control tenant by tenant above the rent floor; voucher redemption and incremental return visits; 30/60/90-day return rate.

Every figure here is a scenario or a sensitivity, not a forecast. Sources: appendix source slides.

How the software and running costs are derived

Work breakdown: person-days (assumed effort)

Work package	Production	Pilot
Member identity and entitlement API	10	10
Booking service and capacity guard	12	12
Deposit, refund, no-show, reconciliation	12	12
LPR ingest, plate ledger, barrier control	15	7
Operator console, incident log, audit	10	5
Security, deployment, monitoring	6	4
Test, UAT, fault drills, commissioning	12	8
Project management and QA, +15%	11.5	8.7
Total person-days	**89**	**67**
× RM623 a day: market value, not charged	**RM55,100**	**RM41,500**

Cross-checks. Vendor pages price a comparable app (payment, admin) at RM50–150k, which includes their margin; the prototype here already exists. Payment integration takes 2–3 weeks per vendor guidance (we use 12 days). Surveys put average software overrun at 30–40%: the high-end range adds 35%. All maintenance, enhancements included, would be 30% of build a year (running RM10.2k).

In-house build: market value, not charged

RM115,701 a year (Payscale, 2026) + 13% employer on-costs (EPF, SOCSO/EIS) ÷ 210 productive days (assumed) = **RM623 a person-day**, no agency margin. Shown for transparency only: SDP is assumed to have the talent.

Upfront cost (capex): cash items, mostly allowances

Item	RM
Parking-system and barrier vendor interface	15,000
Payment gateway onboarding and security test	7,500
Blind-spot CCTV, 4 cameras (RM1,500 each)	6,000
Power, network and works (allowance)	7,500
Guidance display (optional, Viskou list price)	2,033
Upfront total (capex), as on the cost slide	**38,033**

Running cost a year

Item	RM
Maintenance: 15% of third-party build (derived, see note)	3,375
Hosting: 2 × t3.medium, AWS Malaysia, 24×7, at RM4.1/USD	3,400
Running	**6,800**

Effort per work package is an assumption: it is itemised so a vendor can replace each line with a quote. Payscale and AWS figures were opened and checked. Maintenance: studies put it at 60–90% of software cost; at the lowest, 60% over 5 years, that is 30% of build a year, and corrective plus adaptive work (45%) gives 13.6%, rounded up to 15%. Enhancements are excluded. A vendor quote would add its margin on top of this cost basis.

Deployment options: what each spends, earns and risks

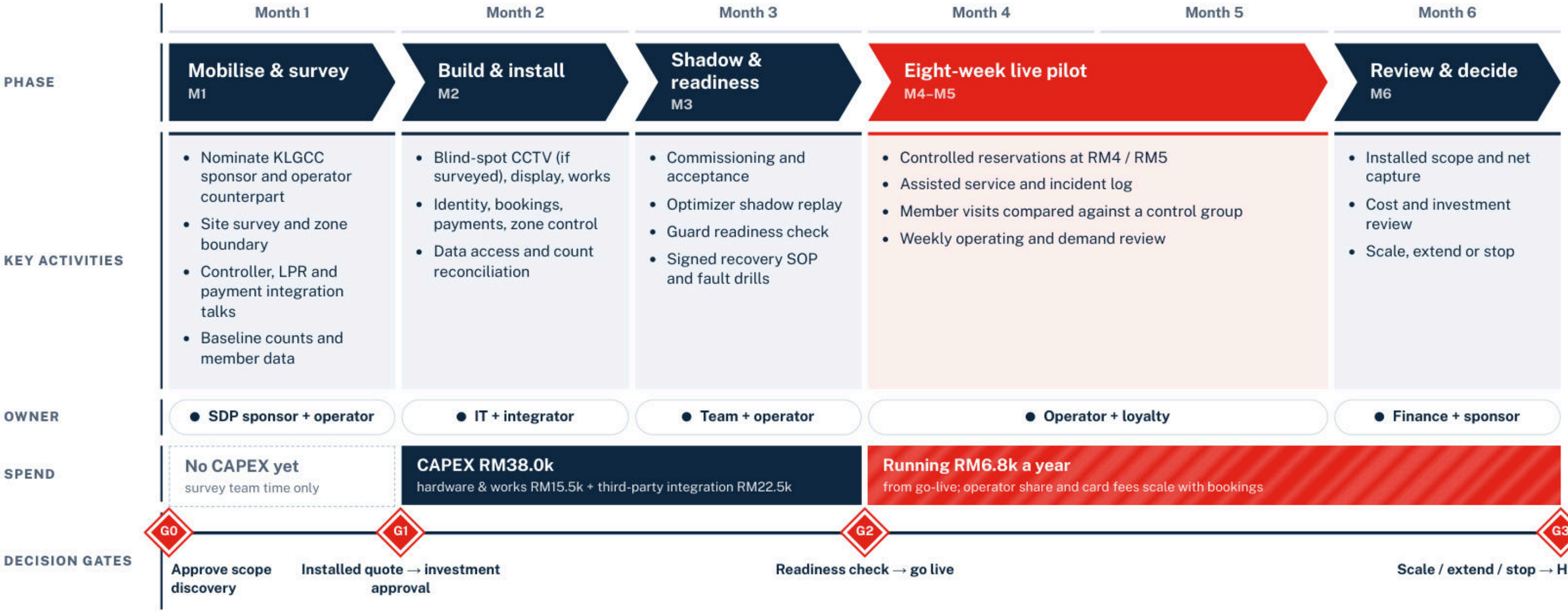
Option	Upfront	Running a year	Bookings a day for 3-year recovery	Payback at 22 / 44 bookings a day	What it adds	Main risk
Existing LPR + software	RM30.0k	RM6.8k	21	2.6 yr / 1.0 yr	Entry and exit plate ledger; the capacity guard and booking checks.	No independent count. At 0.5 to 1% missed reads the 200-bay simulation shows 5 to 6 unseen and 7 to 10 ghost cars (99th percentile).
+ CCTV count audit and display (base case)	RM38.0k	RM6.8k	24	3.4 yr / 1.3 yr	4 cameras over blind spots, one display; an independent count and audit trail.	Coverage depends on the site survey; add cameras only where it finds gaps.
High case: 10 cameras, 35% build overrun	RM54.9k	RM8.0k	32	5.4 yr / 2.0 yr	Fuller aisle coverage.	Revenue is unchanged, so only worth it if the survey finds wide blind spots.
Bay-level sensing or a digital twin	Not costed: no quote or source.				Per-bay occupancy and guidance.	Not tested. The guarantee here is checked at zone level from counts, not per bay.

What it earns is the same under every option: RM3.04 gross per accepted booking at 50% members, 0.735 net of tax, operator share and card fees (slide 7). The options differ in cost and in how well the count can be trusted. Upfront cost is a scenario: third-party items and site works assumed, in-house build not charged, hardware from listed prices.

Revenue is the same under every option; the choice is how much count accuracy to buy.

Derived from v3_costs.py (each input marked sourced, derived or assumed) and v3_zone200.py. Break-even uses the same net factor and gross per booking as the revenue slide. Not a quote.

Implementation roadmap: first six months



SO WHAT Four gates tie spending to evidence: no CAPEX before the installed quote, no paid bookings before readiness, no scale-up before the month-6 review.

Sources: V3 simulation; data register in the appendix. Proposed sequence; months are indicative and sit inside H1 (0–6 months). Phases, owners and gates follow the eight-week pilot schedule and the next-steps slide. Costs are one reference zone before unpriced equipment, central support, project taxes and contingency. Not an agreed project plan or committed budget.

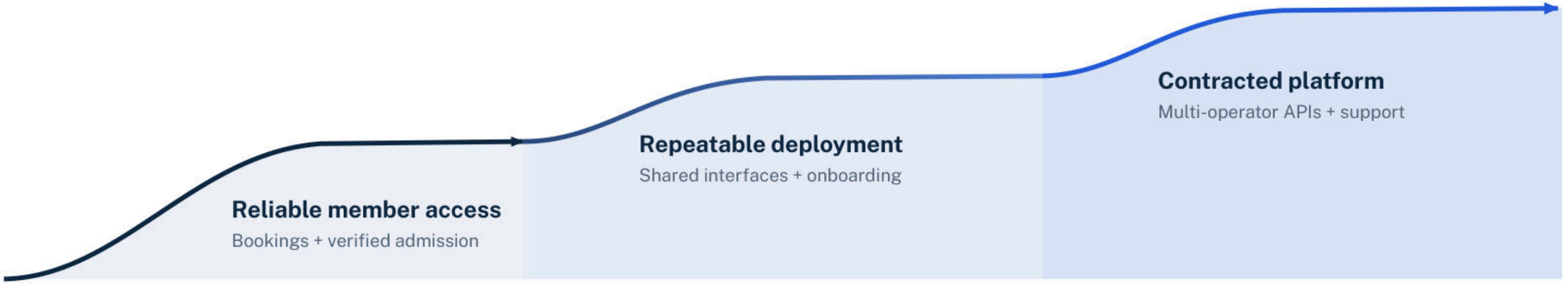
Three-horizon roadmap

Prove the service. Replicate across SDP. Scale to external operators.

H1 | PROVE
KLGCC member service
0–6 months

H2 | REPLICATE
SDP operating system
6–24 months
Potential sites: Elmina + KL East

H3 | SCALE
External platform
24–60 months



Illustrative expansion of scope; planning windows are not a linear time axis

◇ **GATE 1: BEFORE EXPANSION**
Service readiness + full-cost review

◇ **GATE 2: BEFORE WIDER DEPLOYMENT**
Repeatable delivery at approved sites

◇ **GATE 3: BEFORE EXTERNAL ROLLOUT**
Signed contracts + viable support

Source: Three-horizons framework, adapted for ParkFlow. Curves indicate broader proposed scope; dates are indicative focus windows, not a linear time axis or revenue/adoption forecast.

Pilot measures and responsibilities

Value channel	Measure	Attribution / cash boundary	Owner
Paid service	Offer conversion, realized fee and refunds	Randomised price offers; settled net receipts	Member team / Finance
Member retention	30/60/90-day return rate; extra visits/member	Eligible control cohort; baseline-adjusted visits	CRM / analytics
Tenant and mall value	Attributed tenant sales; marginal rent/parking cash	Lease-by-lease capture; displaced visits and fees deducted	Leasing / Finance
Operating efficiency	Incidents, staff minutes, queue and bay use	Observed baseline; actual staff/cost reduction	Parking operator
Incremental ML value	Same-controller ML vs strongest simple forecast	Measured operating difference; do not price the preference score	Product / data team
Investment decision	Complete installed cost and net annual cash	Quoted scope; no double-counting across value channels	Sponsor / procurement

Total net benefit: booking contribution + attributable marginal mall cash + realized operating savings − extra support, incentives and displaced income. Track tenant-sales and member-experience benefits alongside this cash calculation.

Proposed measurement plan, not completed field evidence. External programme and spending benchmarks establish relevance, not ParkFlow retention uplift. Measure uncertainty and avoid counting repeat bookers as saved customers.

Benchmark methods and scope

Evidence layer	Completed work	Boundary
Public occupancy	18,074 forecasts; 28 UK car parks; 30/60/120/180 minutes. Weather and event flags added no accuracy.	Same-site historical time holdout, not KLGCC or booking requests.
Camera perception	Earlier 99.05% on 78,270 held-out-camera patches.	One outdoor site; indoor transfer and false-free counts still need testing.
Earlier controller studies	6,500 synthetic policy-day runs; guard only matched LP; learned forecasts showed no clear advantage.	Synthetic demand; superseded by the commitment study for the architecture claim.
Commitment planner study	Synthetic 100-bay days: settings tuned on 150 days, tested on 200 fresh days; plus 60 random sets of assumptions.	Stay effects and demand shapes are assumed; one level, no fallback modelled.
Commercial model	Published equipment/payment benchmarks; one reconciled V2 scenario and derived recovery thresholds.	No installed quote, local paid demand, controlled retention effect or agreed net capture rate.

Only occupancy and camera results use real data. Demand and booking behaviour stay synthetic until the pilot.

Sources: commitment study; Event-driven benchmark, 8 Nov 2026; earlier batch and occupancy report; camera benchmark. No field safety, retention or commercial return is established.

SDP already earns turnover rent and car park income



SDP Annual Report 2022, page 126: the segment that earns recurring income from rental of investment properties.

FINANCIAL PERFORMANCE

IAM recorded an 18% increase in revenue to RM107.9 million as compared to FY2021, with the retail business and commercial assets as main contributors. The increase in revenue from retail was mainly due to new tenants opening and a higher turnover rent post COVID-19 lockdowns. Revenue from commercial assets was mainly from higher rental income arising from better occupancy and rental rates, further enhanced by increased income from the car parks.

Same section, financial performance (highlights added by us).

SEGMENT REVENUE

+18%

to RM107.9m in FY2022, with the retail business and commercial assets as the main contributors.

TURNOVER RENT

Retail revenue rose on new tenants and **higher turnover rent**. This is the GTO income line a member programme can add to.

CAR PARKS

Commercial-asset income also rose, **helped by car park income**.

FY2022 was a recovery year after the lockdowns. The report shows turnover rent is a real income line for SDP. How much ParkFlow members add is what the pilot measures.

Source: SDP Annual Report 2022, Investment and Asset Management (investment highlights). Excerpts shown as published; the segment also includes other malls, not only KLGCC.

Public sources

Commercial sources, checked 9 Nov 2026

KLGCC Mall: official parking information

Current tariffs, ~1,000 bays and Touch 'n Go card payment; mall LPR not published. New zone needs two additional cameras.

Viskou: DS-TCG406-E ANPR camera

RM3,226 each; equipment-only, tax extra, zero listed stock.

Viskou: DS-TVL224 display

RM2,033 display; equipment-only, tax extra, zero listed stock.

Billplz public rates

Basic: MYR cards 1.8%, wallets 1.5%, B2C FPX RM1.25; approval required.

Billplz fee tax and plan terms

8% SST on gateway fees; Basic has no annual plan charge.

Business value and methods

SDP FY2025 annual report

FY2025: KL East and Elmina achieved 100% committed occupancy; recurring-income context.

Vase.ai: national shopping survey

National survey N=500: RM195 / RM327 frequency-group means; ~RM260 mean. Not causal.

Tropicana Gardens: disclosed lease rates

Tropicana Gardens, 2024 valuation: around 5% for specified trades; 5–15% for other categories; above lease floor.

Paradigm REIT: lease structures

Paradigm PJ/JB: higher-of and percentage-only lease structures. Terms apply by tenant and period.

Technical sources: UCI Parking Birmingham; occupancy / batch report; event-driven report. Other category/privacy references remain on their slides.

Sources: V2 calculation; Sources, scope and unresolved costs. Public list prices and category precedents do not validate installed cost, local retention or willingness to pay. Payment selection must match the actual mix, merchant approval and settlement/refund terms.

Sources: cost model and secondary references

Payscale: senior software developer, Malaysia
Opened: RM115,701 average base, 2026.

AWS pricing, ap-southeast-5 (Malaysia)
Opened: t3.medium USD0.0475 per hour (third-party mirror).

Customs: Guide on Parking Services
Opened: 6% on parking (para 11); reserved-signage charges exempt. 2024 rate hold per PwC and Wolters Kluwer, not opened.

MIDA: statutory contributions
Opened: employer EPF 13% to RM5,000 wage, 12% above; SOCSO 1.75%.

Univ. of Cantabria OCW: software maintenance costs
Opened: maintenance is 60–90% of total software cost across studies.

Moløkken-Østvold and Jørgensen: estimation surveys
Average effort overrun 30–40%, as the authors later summarise it. Not opened.

AdvisoryApps: app cost and payment integration, Malaysia
RM50–150k mid-range app; payment integration 2–3 weeks. Vendor marketing; script-rendered, not verified.

exchangerate.guru: USD to MYR
About 4.06 on 1 Nov 2026 as reported by search; not an official BNM rate.

AlienVoIP: 4CH POE CCTV installation bundle
RM2,599 bundle, RM100 per extra 10 m cable, per search. Script-rendered, not verified.

TP-Link VIGI C340i, Lelong
4MP outdoor IP camera, RM199 listing.

Hikvision 4MP, Lapasar
About RM750 per unit; B2B listing.

Secondary, directional

EdgeProp: KLGCC Mall early footfall
About 100,000 visitors in the first weeks; no annual figure.

Autobuzz: Sunway Pyramid reservation bays
ParkEasy booking reported at 6 (weekday) / 8 (weekend) credits, 1 credit = RM1; 2019, EV bays.

Manitou Springs reservation minutes
One city system: 31% weekly no-show. Directional only.

Campground reservation policy brief
Modelled 10.2% no-show; fees reduce unused bookings.

Car rental no-show analysts
20–30% no-shows, analyst estimate; not parking.

ATRI truck-parking survey
Almost half would refuse to pay for reserved parking; different population.

Right-hand column: found by web search, pages not opened; verify before submission, none measures KLGCC demand or price response. Opened and checked: Payscale, AWS, Customs guide, MIDA, Cantabria notes. The others were not opened or are script-rendered; verify before submission. Effort days, cable run and labour remain assumptions.